INCRELAC™ Scientific Dossier

A Human-Origin GLP-1 Promoting Probiotic & Postbiotic Platform

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Technology Overview

IncreLac™ is a proprietary, human-origin probiotic and postbiotic platform developed to support endogenous GLP-1 signaling through microbiome-mediated mechanisms. The strain has been fully purified, characterized, genome sequenced, and deposited in ATCC, with safety profiling aligned to regulatory expectations.
The technology is positioned to address unmet needs in:

- Metabolic health
- Weight management
- Glucose homeostasis
- Gut-endocrine axis modulation

The Unmet Metabolic Need

Metabolic disorders continue to rise globally, including obesity, insulin resistance, and type 2 diabetes. While pharmacological GLP-1 receptor agonists have demonstrated efficacy, there remains demand for:

- Natural GLP-1–supportive strategies
- Microbiome-based metabolic modulation
- Complementary or adjunct approaches
- Safer, gut-targeted interventions

The gut plays a central role in endocrine regulation. Enteroendocrine L-cells secrete GLP-1 in response to luminal stimuli, including microbial metabolites and specific bacterial interactions.

Innovation Rationale

IncreLac™ was developed to:

- Stimulate endogenous GLP-1 production through gut-mediated mechanisms
- Provide a human-origin, genomically characterized probiotic platform
- Deliver both probiotic and postbiotic bioactivity
- Create a scalable, GMP-manufactured, regulatory-ready ingredient

The technology bridges microbiome science, metabolic endocrinology, and translational product development.

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TECHNOLOGY CHARACTERIZATION

Strain Origin & Purification

- Human-origin isolate
- Carefully purified from donor-derived microbiome source
- Single-strain isolation and stabilization
- Identity confirmed via phenotypic and genomic analysis

The strain is:

- Gram-positive
- Catalase-negative
- Exhibits core probiotic functional attributes

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Safety Attributes

Human-origin strain

- Non-pathogenic
- No hemolytic activity
- Controlled manufacturing process
- Stability validated under storage conditions

ATCC Deposition

The strain has been deposited in the American Type Culture Collection (ATCC), ensuring:

- Identity traceability
- Intellectual property security
- Regulatory transparency
- Research reproducibility

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MECHANISTIC EVIDENCE

GLP-1 Stimulation in Human Intestinal Cell Models

In vitro studies using human intestinal cell lines demonstrate:

- Increased GLP-1 secretion upon exposure to IncreLac™
- Dose-dependent stimulation
- Statistical significance versus control
- Reproducibility across biological replicates

Data suggest direct or indirect interaction with enteroendocrine pathways.

Proposed Gut–Endocrine Mechanisms

Potential mechanisms under investigation include:

- Microbial metabolite production influencing L-cell activation
- Short-chain fatty acid modulation
- Gut barrier interaction and improved epithelial signaling
- Interaction with enteroendocrine receptors

Further validation studies are ongoing to clarify signaling cascades.

Postbiotic Activity

In addition to live-cell effects, IncreLac™–derived postbiotic fractions demonstrate:

- Bioactive components capable of modulating GLP-1 signaling
- Stability advantages for formulation flexibility
- Potential heat-stable activity

This dual probiotic + postbiotic platform enhances commercial versatility.

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DEVELOPMENT & COMPLIANCE

GMP Scale-Up

IncreLac™ is currently:

- Scaled in GMP-certified manufacturing facilities
- Produced under controlled fermentation conditions
- Batch-tested for purity and potency
- Stabilized for human consumption formats

Quality control includes:

- Identity verification
- Viability assessment
- Microbial purity testing
- Stability profiling

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GRAS Pathway

Preparation underway for:

- GRAS self-affirmation
- Dossier compilation
- Toxicological safety evaluation
- Expert panel review

Documentation includes:

- Genomic safety data
- Manufacturing process controls
- Stability and shelf-life data

Regulatory Alignment Strategy

The development program aligns with:

- FDA dietary supplement framework
- GRAS requirements
- Structure/function claim substantiation
- Clinical substantiation pathways

Future regulatory positioning may include:

- Medical food
- Adjunct metabolic support
- Combination therapeutic research

Planned Clinical Endpoints

Upcoming clinical trials aim to evaluate:

- Fasting GLP-1 levels
- Postprandial GLP-1 response
- Glucose regulation markers
- Insulin sensitivity indices
- Body weight and composition parameters
- Gut microbiome modulation

Study design will include randomized, placebo-controlled frameworks.

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COMMERCIAL PATHWAYS

Target Markets

- Metabolic health supplements
- Weight management formulations
- Blood sugar support products
- Functional foods and beverages
- Synbiotic blends

Weight Management Support

As a microbiome-based GLP-1 promoter, IncreLac™ may serve as:

- Natural adjunct to GLP-1 pharmaceuticals
- Standalone metabolic support ingredient
- Preventive metabolic wellness product

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Combination Therapies

Potential combination opportunities include:

- Fiber blends
- Prebiotics
- Polyphenols
- Protein-based metabolic products
- GLP-1 receptor agonist adjunct formulations

Co-Development Models

MusB Research offers flexible collaboration models:

- Licensing agreements
- Co-branded development
- Exclusive distribution partnerships
- Joint clinical trial sponsorship
- Data-sharing partnerships

Strategic Value Proposition

IncreLac™ delivers:

- Human-origin scientific credibility
- Genomic transparency
- Regulatory readiness
- Mechanistic evidence
- Scalable manufacturing
- Market differentiation

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INCRELAC™

IncreLac™ represents a next-generation microbiome platform engineered to bridge metabolic science and commercial opportunity. Through rigorous characterization, regulatory preparation, and translational validation, MusB Research provides sponsors with a foundation for responsible innovation and competitive differentiation.